DYLAN CHEN

dylanchen4306@gmail.com |929-488-9693 | Linkedin | Portfolio

EDUCATION

Northeastern University - *Boston, MA* *Expected May 2028*
Bachelors of Science in Computer Engineering and Computer Science *GPA: 3.5/4.0*
Relevant Coursework: Algorithms and Data, Digital Design/Computer Organization, Networks, Analysis of Random Phenomena, Embedded Systems, Circuits and Signals, Intro to Databases, Computing Fundamentals
Activities & Leadership: UX Designer *(Sandbox NU)*, Software Developer *(Oasis Accelerator)*

SKILLS

- **Hardware & Prototyping:** SolidWorks (CAD), KiCad (PCB Layout), 3D Printing, Oscilloscope, DMM, Soldering
- **Embedded Systems & Firmware:** C/C++, ESP32, Arduino, SystemVerilog, RISC-V, Sensors, I2C, SPI, UART, PWM, ADC
- **Software & Tools:** Python, Java, Kotlin, MATLAB, Git, CI/CD, System Integration, Imaging/Computer Vision
- **Interface & Design (UX):** HTML/CSS, JavaScript/TypeScript, Blender, Figma, Adobe Creative Suite

TECHNICAL EXPERIENCE

Product Test Engineer Co-op - *SharkNinja* *January 2026 - June 2026*

- Fabricated custom electromechanical validation fixtures using Solidworks and Arduino (C++) with I2C to automate manual life cycle and material testing for consumer products, increasing daily throughput of test cycles by 600%.
- Developed a Python automation script to automatically parse CSV data from a custom pressure testing rig for 180+ water filter test cycles, formatting data into deliverables and reducing turnaround time by 50%.
- Engineered a Python-based timelapse analysis tool utilizing an Otsu threshold computer vision algorithm and image calibration, cutting post-processing analysis time by 98%.
- Refined hardware testing protocols and tooling tolerances, scheduling weekly 1-on-1 meetings with managers and used cross-functional team feedback to increase accuracy of data and standardize quality assurance metrics.
- Directed cross-team training for project managers, directors, and non-technical staff on custom software tools and authored standardized operating procedures (SOPs) for complex quality test protocols.

PROJECTS

4-Degree-of-Freedom (4DOF) Robotic Arm July 2026 - Present

- Design and prototype a 4-DOF servo-driven robotic arm, using 3d printed components modelled in Solidworks and housing an ESP32 based wireless control architecture.
- Develop a full-stack web interface with Javascript, HTML, CSS, and Three.JS rendering for a future control hub GUI and inverse kinematics + PID motor management

Smart Control Knob (NOB) *September 2025 - Present*

- Developed C++ firmware on an ESP32-C3 that uses interrupt-driven decoding to read a rotary encoder with built-in contact-bounce rejection, and drives a round SPI display that shows live volume levels.
- Built a Python host app that receives knob input over serial (UART) and controls Windows system volume in real time, creating a working hardware-to-PC audio control loop.

Electrocardiogram System (ECG) *November 2025 - December 2025*

- Designed and built a full analog-to-digital ECG system using an AD627 instrumentation amplifier and cascaded active high- and low-pass filters to remove DC offset, reduce noise, and prevent aliasing before digitization.
- Processed digitized ECG signal in MATLAB with Butterworth and Parks-McClellan filters to calculate heart rate, and verified amplifier performance against datasheet specifications.

INTERESTS

UIUX, Photographing wallpapers, Coffee brewing, Baking without measurements, Climbing rocks, doodles, and noodles